



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

SECOND YEAR END OF YEAR EXAMINATIONS FOR THE DEGREE OF BACHELOR OF MEDICINE AND SURGERY (MBChB) WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

MBS 200: HUMAN ANATOMY II

Date: 21st November, 2017

Time: 2.00 - 5.00pm

INSTRUCTIONS:

- Do not write your name on any paper you use
- Section A Multiple Choice Questions (MCQ)- 40 mks
- Section B Fill in the blanks 10 mks
- Section C: Cross Matching 10 mks
- Section D: Short Answer Questions (SAQ) 30 mks
- Section E: Diagram 30 mks
- Answer ALL the Questions
- Read very carefully the additional instructions preceding each section
- The paper has a total of 130 marks and 180 minutes
- Do not answer any Question in the Question paper.

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ISO 9001:2008 CERTIFIED



SECTION A, MCQ TYPE 3

i) There are 60 questions in this section. (60 minutes/60 marks)

ii) Choose the most appropriate answer and mark it with (x) in the answer sheet provided.
Mark the best answer

1. The perineum
 - A. is all of the soft tissue enclosing pelvic inlet.
 - B. Can be divided into urogenital and anal region.
 - C. ends anteriorly at the external reproductive organs
 - D. ends posteriorly at the anus.
2. The hiatus of the pelvic diaphragm
 - A. is a cleft between the anteromedial margin of pelvic diaphragm.
 - B. is filled with the urogenital diaphragm.
 - C. is enclosed by superior and inferior fascia of urogenital diaphragm.
 - D. is the main channel connecting the pelvic spaces and perineal space.
3. The prostate
 - A. is located between the apex of bladder and urogenital diaphragm.
 - B. is placed in front of the uterus.
 - C. has venous plexus between the prostatic capsule and sheath.
 - D. can not be touched during check-up.
4. Anal canal
 - A. continues with the rectum at the level of pelvic diaphragm.
 - B. terminates at the dentate line.
 - C. is about 7~8 cm in length.
 - D. has two flexures in sagittal plane.
5. With regard to the urethra in the young adult male, which is NOT true?
 - A. the prostatic urethra is narrower than the membranous urethra
 - B. it is approximately 20cm long
 - C. has a short-dilated region just proximal to the external urethral meatus d) it is horizontal in cross-section when empty
 - D. the bulbous part is part of the spongy urethra
6. The testis:
 - A. the epididymis arises from the lower pole of the testis
 - B. the tunica vaginalis surrounds the testis but doe not encase the epididymis
 - C. vas deferens ascends on the anterior surface of the testis
 - D. the testicular appendix lies on the testis upper pole near the origin of the epididymis
7. The testicular blood supply:
 - A. is mainly from the ductal artery

- B. the right drains directly into the inferior vena cava
 - C. venous drainage does not have valves
 - D. a varicocele is more common on the right than the left
8. Which of the following positions of the uterus & vagina is the "usual" position in a female?
- A. Anteverted & retroflexed
 - B. Anteverted & anteflexed
 - C. Retroverted & retroflexed
 - D. Retroverted
9. Which of the following muscles is most important in creating the pelvic diaphragm?
- A. Piriformis
 - B. Obturator internus
 - C. External anal sphincter
 - D. Levator ani
10. Which structure is found only in males?
- A. Anterior recess of ischioanal fossa
 - B. Genital Hiatus
 - C. Ischiocavernosus muscle
 - D. Rectovesical pouch
11. Which structure does NOT form part of the boundary defining the trigone of the bladder?
- A. Interuterine crest
 - B. Left Ureteric Orifice
 - C. Right Ureteric Orifice
 - D. Urachus
12. Which of the following is considered a part of the broad ligament of the uterus ?
- A. Mesovarium
 - B. Ovarian ligament
 - C. Round ligament of the uterus
 - D. Suspensory Ligament of the ovary
13. The prostate gland:
- A. Contains upper, middle and lower lobes
 - B. Encircles the urethra
 - C. Is well imaged radiologically using an intravenous urogram
 - D. B and D
14. Which structure is NOT found within the true pelvis?
- A. Femoral nerve
 - B. Hypogastric nerve
 - C. Internal pudendal artery
 - D. Obturator artery
15. Which of the following is true about the male pelvis:
- A. larger pelvic inlet
 - B. smaller subpubic angle
 - C. straighter sacral curvature
 - D. larger pelvic outlet
16. The inner membranous walls of the vagina that form folds are called:
- A. Fornices.

- B. Mons pubis.
 - C. Vestibule.
 - D. Labia Majora
17. At which month do the testes begin to descend into the scrotum of the fetus?
- A. 4
 - B. 5
 - C. 6
 - D. 8
18. Which of the following is a retroperitoneal structure:
- A. Uterus
 - B. Liver
 - C. Kidney
 - D. Appendix
19. Parts of the intestines with a mesentery include:
- A. Jejunal and rectum
 - B. Jejunum and ileum
 - C. Jejunum and ascending colon
 - D. Ascending and descending colon
20. Which of the following is a ligament of the liver:
- A. Median umbilical folds
 - B. The ligamentum Venosum
 - C. Median umbilical ligament
 - D. None of the above
21. Transpyloric Plane lies at the level of:
- A T12
 - B L1
 - C L2
 - D L3
22. The platysma is innervated by the:
- A. Mandibular nerve
 - B. Maxillary nerve
 - C. Accessory nerve
 - D. Cervical branch of the facial nerve
23. The following skull bones are derived from the first arch except:
- A. Mandible
 - B. Maxilla
 - C. Zygoma
 - D. Styloid process of the temporal bone

24. The layer of the scalp which allows movement to take place is the:
- A. Skin
 - B. Connective tissue.
 - C. Aponeurosis
 - D. Loose areolar tissue.
25. Cerebral spinal fluid is found in:
- A. Superior sagittal sinus.
 - B. Subdural space
 - C. Transverse sinus
 - D. Subarachnoid space
26. During the 4th week of development, in what region does the pharyngeal apparatus develop?
- A. Cephalic
 - B. Foregut
 - C. Midgut
 - D. Hindgut
27. What type of cells makes up the pharyngeal apparatus pouches?
- A. Endoderm
 - B. Ectoderm
 - C. Mesenchyme
 - D. Intermediate mesoderm
28. What type of cells makes up the pharyngeal apparatus clefts?
- A. Endoderm
 - B. Ectoderm
 - C. Mesenchyme
 - D. Intermediate mesoderm
29. What pharyngeal/branchial arch does the oropharyngeal membrane develop from?
- A. Sixth
 - B. Fourth
 - C. Third
 - D. Second
-
30. A typical pharyngeal/branchial arch contains all of the following EXCEPT:
- A. Nerve
 - B. Artery
 - C. Vein
 - D. Somitomeres
31. What two structures only form on the ventral side of the bile duct?
- A. Liver; Pancreas
 - B. Liver; Gallbladder
 - C. Pancreas; Gallbladder
 - D. Heart; Liver
32. Which of the following is NOT a part of the developing pancreas?
- A. Dorsal bud

- B. Ventral bud
 - C. Ventral bile duct
 - D. Main pancreatic duct
33. The annulus fibrosus is formed from the:
- A. Notocord
 - B. Dermomyotome
 - C. Sclerotome
 - D. Mesoderm
34. A tuft of hair on the back of a newborn is likely associated with:
- A. Spina bifida occulta
 - B. Klippel-Feil syndrome
 - C. Spina bifida cystica (myelomeningocele)
 - D. Hypertrichosis lanuginosa
35. Which of the following is lined by a serosa?
- A. Genitourinary tract
 - B. Peritoneal cavity
 - C. Respiratory tract
 - D. Alimentary canal
36. What type of tissue lines the esophagus?
- A. Simple squamous epithelium
 - B. Simple cuboidal epithelium
 - C. Simple columnar epithelium
 - D. Stratified squamous epithelium
37. Which of the following belongs exclusively to the conducting portion of the respiratory system?
- A. Alveolar pore
 - B. Alveolar sac
 - C. Alveolar duct
 - D. Respiratory bronchiole
38. Which of the following cell types is NOT found in respiratory epithelium?
- A. Ciliated cells
 - B. Sensory cells
 - C. Goblet cells
 - D. Satellite cells
39. Plicae circularis (valves of Kerkring) are present in
- A. stomach
 - B. ileum
 - C. duodenum
 - D. jejunum
 - E. all but the stomach
40. The epithelium of the gallbladder is composed of
- A. simple columnar epithelium
 - B. psuedostratified epithelium
 - C. ciliated columnar epithelium
 - D. stratified squamous epithelium
 - E. simple squamous epithelium

SECTION B: fill in the blanks (10 Marks, 10 Minutes)

1. The esophagus is lined with _____ epithelium.
 2. The _____ muscle has spindle shaped fibers but not striated.
 3. The _____ cells cannot take part in impulse transmission in the nervous system.
 4. The Esophagus Starts at C6 vertebra and ends at _____ vertebra.
 5. The stomach appears as a fusiform dilation of the foregut in the _____ week of development
 6. The hepatic tract in the liver acini is has a centrally located _____ vein
 7. The _____ is produced by type II alveolar epithelial cells.
 8. The pelvic brim is also formed by the Promontory _____ of the sacrum.
 9. Ischial tuberosities laterally form the _____ outlet.
 10. The sacrococcygeal joint is a _____ joint.
-

SECTION C: Match the following with their correct answers in table B (10 Marks/ 10 minutes)

Match the structures in column A with those in column B (Ensure you clearly state the answer against each number ie if number 8 is Waldeyer's fascia, then state 8 = A)

Column A	Answer	Column B
1. Duodenum		A. Waldeyer's fascia
2. Appendix		B. Angular incisure
3. Anus		C. hepatopancreatic duct
4. Ampulla of Vater		D. 3 or more arterial arcades
5. Jejunum		E. Pre-pyloric vein of Mayo
		F. Celiac trunk
		G. Lymphoid tissues
		H. Vagus nerve
		I. Morgagni Columns
		J. Long vasa recti fewer arcades
		K. Brunner's glands
		L. Sacculations
		M. Taenia

SECTION D: SHORT ANSWER QUESTIONS (40 Marks total, 40 Minutes)

1. Discuss briefly the histological layers and contents of the wall of the GIT tract (10 Marks)
2. Name the structures forming the bed of the stomach (5 Marks)
3. Name the 5 layers of the scrotum (5 Marks)
4. Discuss the formation, circulation and absorption of cerebrospinal fluid (10 Marks)
5. List at least 5 congenital heart defects (5 Marks)
6. List the structures (Ligaments and muscles) that provide anatomical support to the uterus (5 Marks)

SECTION E; DIAGRAMS (30 Marks)

1) PLATE A; LABEL THE STRUCTURES MARKED 1 TO 10 IN THE GIVEN DIAGRAM (10 Marks, 10 Minutes)

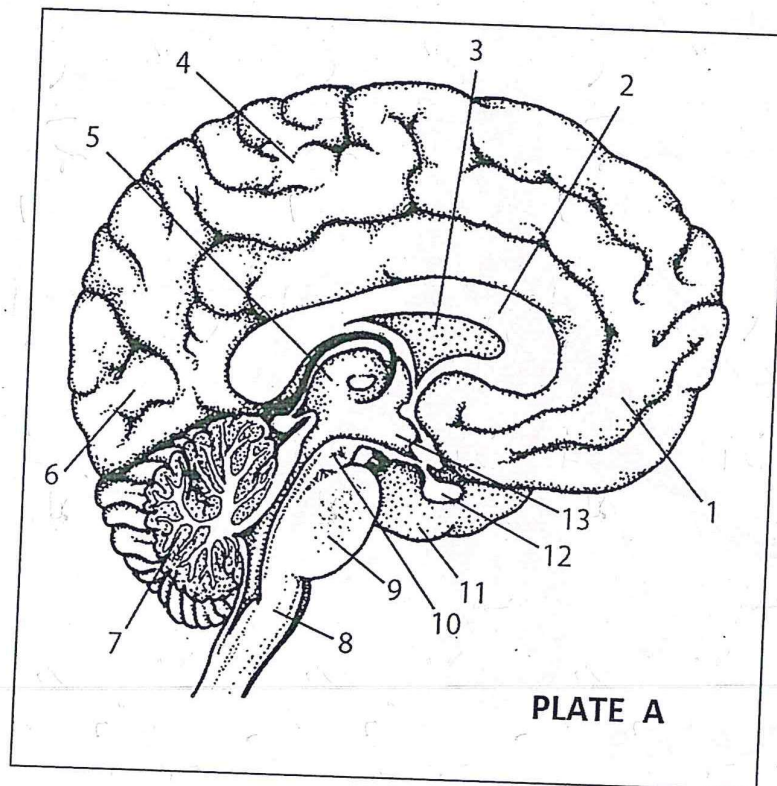


PLATE B: LABEL THE MARKED STRUCTURES A-J IN THE GIVEN DIAGRAM (10 Marks, 10 Minutes)

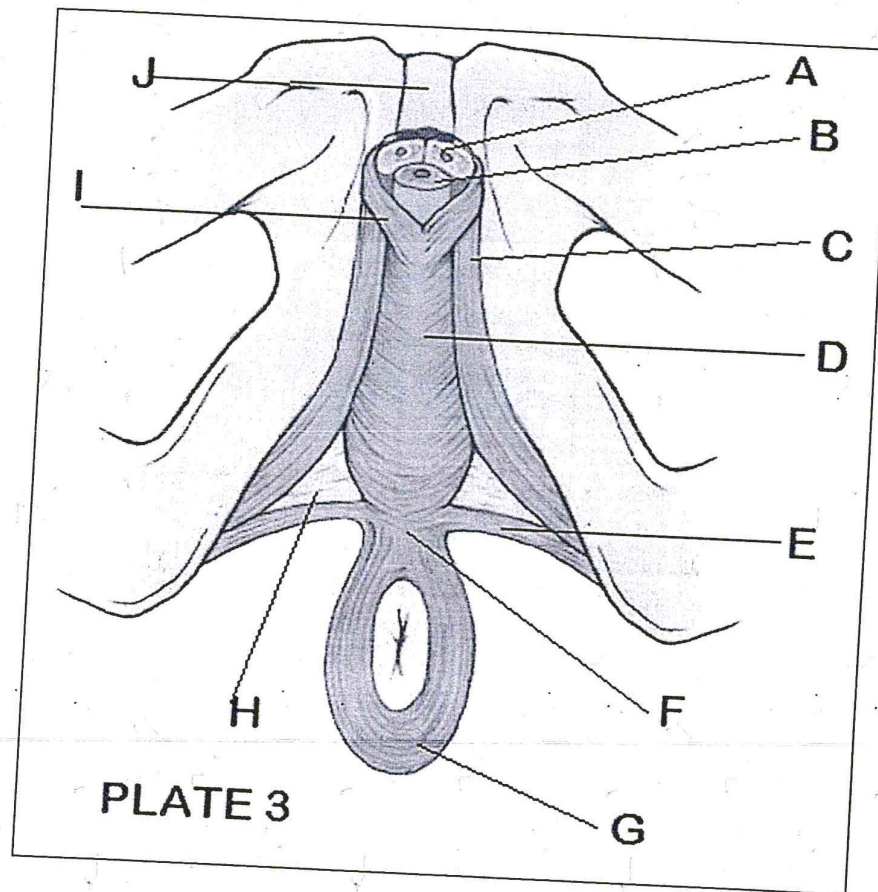
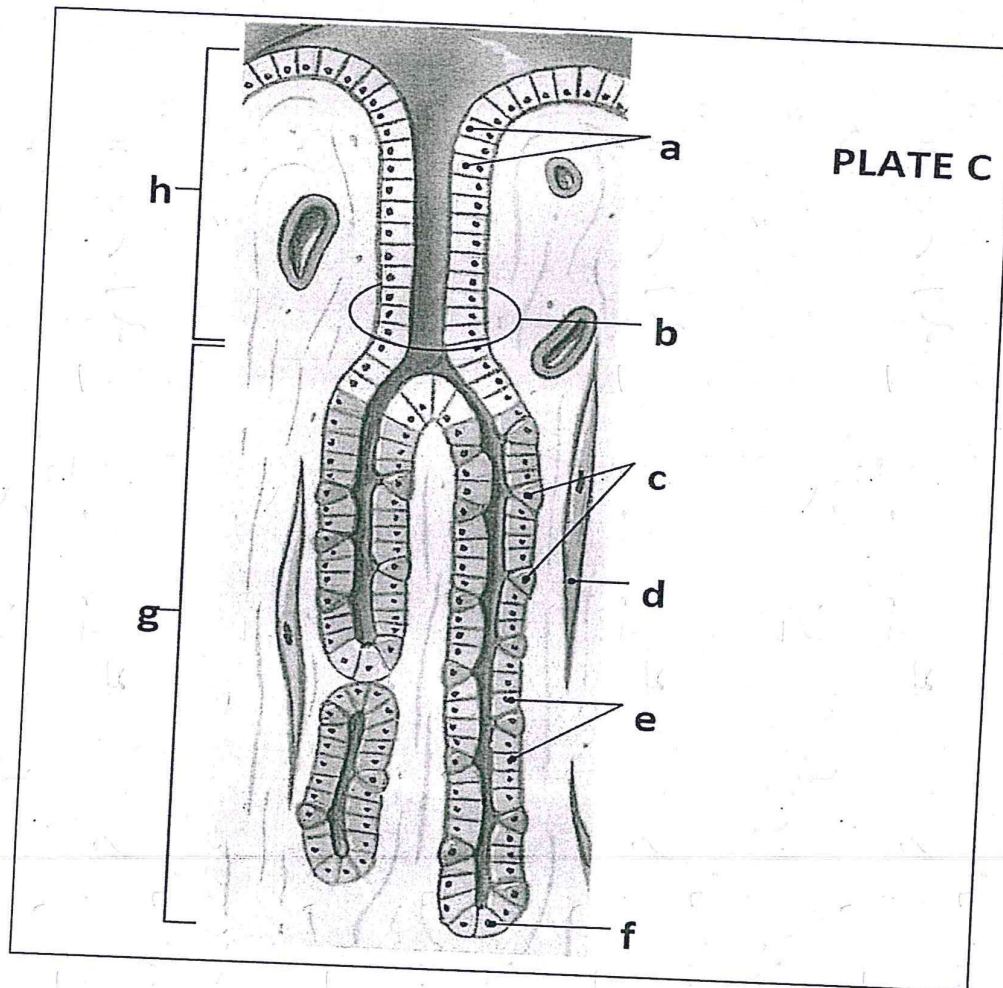


PLATE C: (10 Marks, 10 Minutes)

- I) Identify the structure and where it is located (2 Marks)
II) Label the marked structures A-H in the given diagram (8 Marks)





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INFORMATION TECHNOLOGY**

MAIN CAMPUS

MBS 201: MEDICAL PHYSIOLOGY II

Date: 27th November, 2017

Time: 2.00 - 5.00 pm

INSTRUCTIONS:

- SEE INSIDE
- This paper consists of FOURTEEN (14) printed pages



THIS PAPER CONSISTS OF THREE SECTIONS A, B & C.
ANSWER ALL QUESTIONS FROM EACH SECTION

SECTION A: MULTIPLE CHOICE QUESTIONS (75MIN)
ONE CORRECT ANSWER

1. All the following are correct concerning cardiac muscle EXCEPT:
 - A. Cardiac cells are smaller than skeletal muscle cell
 - B. Electrical communication between cardiac cells is maintained via gap junctions which are specialized portions of the intercalated discs
 - C. Mechanical attachment of cardiac cells is at the intercalated disc.
 - D. Transverse tubules are larger in skeletal muscle than in heart muscle allowing more diffusion of Ca^{2+} into the interior of the cell.
2. Which of the following is usually the dominant pacemaker?
 - A. SA node.
 - B. AV node
 - C. His bundle.
 - D. Purkinje fibers.
3. All the following are true concerning the parasympathetic system control of the heart EXCEPT:
 - A. It affects muscarinic receptors.
 - B. It decreases heart rate via the SA node.
 - C. It decreases conduction velocity via the AV node.
 - D. It can be blocked by beta blockers.
4. Which of the following is a TRUE statement concerning pacemaker potential?
 - A. Specialized cells depolarize during phase 4, but ventricular and atrial muscle cells do not.
 - B. Specialized cells depolarize during phase 3, but ventricular and atrial muscle cells do not.
 - C. Specialized cells repolarize during phase 4, but ventricular and atrial muscle cells do not.
 - D. None are correct
5. Which of the following is the property of a cardiac cell to initiate and fire an action potential on its own without external stimulation?
 - A. Selectivity.
 - B. Spontaneity.
 - C. Automaticity.
 - D. Conductance.
6. Which of the following does NOT show rapid initial depolarization at the start of an action potential?
 - A. SA node.
 - B. Atrial muscle.
 - C. Purkinje fibers.
 - D. Bundle of His.
7. Sodium enters the cell during the upstroke of the action potential. What is the major mechanism for removing Na from the cell?
 - A. It passively diffuses out of the cell.

- B. It is extruded via an ATP-dependent Na-K pump.
 - C. It is extruded via an ATP-dependent Na-Ca pump.
 - D. It is extruded via an ATP-independent Na-K pump.
8. Which of the following is INCORRECT concerning diastolic depolarization at the SA node?
- A. It results from a decrease in I_K .
 - B. Results from the activation of I_f carried mainly by Na ions.
 - C. Its rate is decreased by sympathetic stimulation.
 - D. Its rate is decreased by decreasing I_{Ca}
9. The following are correct concerning local circuit flow in the heart EXCEPT?
- A. It flows from regions with high membrane potential to regions with low potential.
 - B. The cell with the more negative potential is called the current source, while the one with the more positive potential is called the current sink.
 - C. The circuit flows from the current source to the current sink.
 - D. The circuit flows from the active cell to the resting cells in its vicinity.
10. Which of the following pairs is INCORRECT?
- A. P wave: atrial depolarization
 - B. QRS complex: ventricular depolarization
 - C. T wave: ventricular repolarization
 - D. QT interval: Measure of duration of atrial action potential
11. 37. Which of the following pairs is INCORRECT concerning the Einthoven triangle?
- A. Lead I: RA/LA
 - B. Lead II: RA/LL
 - C. Lead III: LA/LL
 - D. All of the pairs are correct.
12. The following are affected by the preload in the heart muscle EXCEPT?
- A. End systolic volume
 - B. End diastolic volume
 - C. Stroke Volume
 - D. Ejection fraction.
13. Which of the following is INCORRECT concerning isotonic contraction in the heart?
- A. One occurs during the ejection phase of the cardiac cycle.
 - B. Is typically the first step in heart muscle contraction followed by an isometric contraction.
 - C. It involves a change in muscle length against a constant load.
 - D. An increased afterload will slow the velocity of muscle shortening.
14. Which of the following is NOT an effect of a positive inotropic agent on the heart?
- A. It increases stroke volume.
 - B. It increases the initial velocity of muscle shortening at all loads.
 - C. It increases the rate of blood ejection from the heart.
 - D. It decreases the rate of rise in ventricular blood pressure during systole.
15. Which of the following will require the smallest increase in oxygen consumption by the heart?
- A. Generating an increase in left ventricular pressure.
 - B. Severe hypertension.
 - C. Generating an increase in stroke volume.
 - D. Aortic stenosis.

16. Which of the following vessels has the largest effect on total peripheral resistance?
 - A. Arteries.
 - B. Arterioles.
 - C. Veins.
 - D. Capillaries.
17. If you decrease a blood vessel's radius by half, by what fraction does the blood flow change?
 - A. $1/2$
 - B. $1/4$
 - C. $1/8$
 - D. $1/16$
18. The following are correct concerning turbulent blood flow EXCEPT?
 - A. The resistance to turbulent flow is greater than laminar flow.
 - B. Turbulence can create vibrations that can be detected as murmurs.
 - C. Turbulence is found normally in the aorta and in narrowed vessels.
 - D. Turbulent flow rate is proportional to the cube root of the driving pressure.
19. Which of the following pairs is INCORRECT?
 - A. Nitric oxide (NO): vasodilator
 - B. Endothelin: vasoconstrictor
 - C. Nitric Oxide (NO): promotes cell growth
 - D. Endothelin: potent positive inotropic effect
20. Which of the following will NOT increase aortic systolic blood pressure?
 - A. Decrease in arterial compliance
 - B. Decrease in aortic distensibility.
 - C. Increase in stroke volume.
 - D. Decrease in ejection velocity.
21. Which of the following is a correct formula for the mean arterial blood pressure?
 - A. $MAP = CO \times SV$
 - B. $MAP = CO \times HR$
 - C. $MAP = SV \times HR \times TPR$
 - D. $MAP = HR \times TPR$
22. Which of the following pairs is INCORRECT concerning arteriolar tone?
 - A. Myogenic control: vascular smooth muscle responds to changes in pressure.
 - B. Metabolic control: accumulation of waste products stimulates vasoconstriction.
 - C. Neural control: vasoconstriction activated by sympathetic nerves.
 - D. Hormonal control: catecholamines affect the adrenergic receptors of the vascular smooth muscle.
23. Which pair is INCORRECT concerning the microcirculation at the capillary level?
 - A. Tight junctions: brain
 - B. Gaps large enough to exchange plasma proteins: kidneys
 - C. Intercellular junctions / pores: exchange of water and lipid-insoluble substances.
 - D. Pinocytosis: cytoplasmic vesicles exchanging fluid and solutes.
 - E. Intracellular fenestrations: exchange of large molecules.
24. If a patient presents with swollen hands, feet, and face, all of the following could be a cause, EXCEPT?
 - A. Large decrease in arterial blood pressure.
 - B. Lymphatic blockage.
 - C. A decrease in plasma proteins.

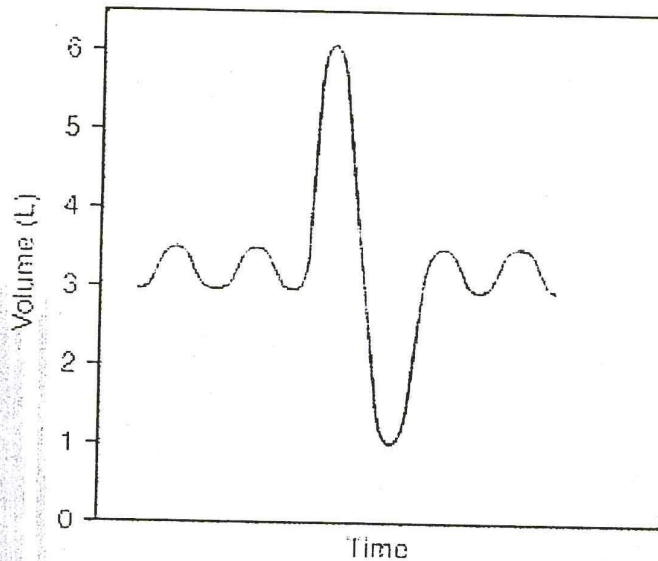
- D. Increase in capillary permeability to proteins.
 - E. Vasodilation.
25. Which of the following is NOT a mechanism to promote venous return and reduce blood pooling?
- A. One way venous valves.
 - B. Parasympathetic stimulation.
 - C. Skeletal muscle pump.
 - D. Thoraco-abdominal pump
26. Which of the following is INCORRECT concerning a patient with ventricular tachycardia?
- A. It occurs when the heart rate drops to dangerously low levels.
 - B. Cardiac output can falls to dangerous levels.
 - C. Arterial blood pressure falls to dangerous levels.
 - D. It involves impaired ventricular filling.
27. Which of the following would NOT occur following a major blood loss?
- A. Vagal tone to the SA node would be decreased which would then increase the heart rate.
 - B. An increase in sympathetic output in arterioles would increase the total peripheral resistance, except in the brain and heart.
 - C. Sympathetic output to the atria and ventricles would increase resulting in an increase in stroke volume.
 - D. Stroke volume would increase while heart rate decreased. Together this would increase the cardiac output.
28. The spread of cardiac excitation is delayed by about 0.1 s at the atrio-ventricular node:
- A. True
 - B. False
29. The second heart sound is due to closure of the aortic and pulmonary valves.
- A. True
 - B. False
30. Which of the following statements concerning the cardiac output is correct?
- A. Is increased by aldosterone released from the adrenal medulla.
 - B. Is increased by stimulation of the vagus nerve
 - C. Can be measured by dividing the oxygen consumption by the difference in PO_2 of the venous and arterial blood.
 - D. Is largely determined by the end-diastolic volume
31. Accessory digestive organs include:
- a. Pharynx
 - b. Gallbladder
 - c. Esophagus
 - d. Duodenum
32. Damage to _____ can lead impairment of the nervous control of GIT
- a. Mucosa
 - b. Muscularis
 - c. Submucosa
 - d. Serosa
33. Digestive enzymes are secreted and released into all these parts EXCEPT:
- a. The mouth
 - b. The small intestines

- c. The stomach
 - d. The esophagus
34. An individual lacking _____ will have pepsin missing from his gastric juice.
- a. Goblet cells
 - b. Parietal cells
 - c. Chief cells
 - d. Enterochromaffin-like cells
35. Damage to the fundus would mainly affect the secretion of:
- a. Pepsinogen
 - b. Histamine
 - c. Gastrin
 - d. Mucus
36. Regarding chyme entering the small intestine
- a. It is basic
 - b. It is buffered by pancreatic bicarbonate secretion
 - c. It has not undergone any digestive process
 - d. It passes through the cardiac sphincter
37. Damage to which of the following would affect fat but not protein or carbohydrate absorption:
- a. The Microvilli
 - b. Brunner's gland
 - c. Lacteals
 - d. The goblet cells
38. Diarrhea may occur as a result of:
- a. Decreased sodium ion reabsorption by the large intestine
 - b. Decreased osmolarity of chyme entering the large intestine
 - c. Damage to the intestinal mucosa due to bacterial enterotoxins
 - d. Active transport of sodium chloride due to gluten consumption
39. Bile:
- a. Is synthesized by the gall bladder
 - b. Is stored in the hepatic sinusoids
 - c. Is solely comprised of bile salts
 - d. Is required for maximal fat digestion and absorption
40. Regarding the pancreas:
- a. It is resistant to proteolytic digestion because of pancreatic trypsin inhibitor
 - b. It secretes bicarbonate to buffer the chyme
 - c. It secretes enzymes required for digestion of all macromolecules
 - d. All of the above
41. Cholecystokinin secretion inhibition causes:
- a. Stimulation of gastric motility
 - b. Stimulation of gallbladder contraction
 - c. Stimulation of pancreatic enzyme secretion
 - d. Stimulation of HCl secretion by parietal cells
42. Gastric acid secretion during the intestinal phase:
- a. Is stimulated by vagal stimulation
 - b. Is inhibited by fat in the chyme
 - c. Is stimulated by gastrin
 - d. Is stimulated by amino acids in the stomach
43. Cholesterol removal and degradation is the primary function of the:

- a. Chylomicrons
 - b. High-density lipoproteins
 - c. Low-density lipoproteins
 - d. Very-low-density lipoproteins
44. Which of the following is a brush border enzyme:
- a. Sucrase
 - b. Aminopeptidase
 - c. Carboxypeptidase
 - d. Maltase
45. How does the autonomic nervous system affect salivary secretory rate:
- a. It is increased in response to parasympathetic and sympathetic stimulation
 - b. It is decreased in response to parasympathetic and sympathetic stimulation
 - c. It is increased in response to parasympathetic stimulation and decreased in response to sympathetic stimulation
 - d. It is decreased in response to parasympathetic stimulation and increased in response to sympathetic stimulation
46. Regarding Gastrointestinal hormones:
- a. They are all steroids
 - b. They all act through secondary messenger molecules
 - c. They are trophic
 - d. Their secretion is regulated by hypothalamic releasing hormones
47. A healthy, 45-year-old man is reading the newspaper. Which of the following muscles are used for quiet breathing?
- a. Diaphragm and external intercostals
 - b. Diaphragm and internal intercostals
 - c. Diaphragm only
 - d. Internal intercostals and abdominal recti
 - e. Scaleni
 - f. Sternocleidomastoid muscles
48. A healthy, 25-year-old medical student participates in a 10-km charity run for the American Heart Association. Which of the following muscles does the student use (contract) during expiration?
- a. Diaphragm and external intercostals
 - b. Diaphragm and internal intercostals
 - c. Diaphragm only
 - d. Internal intercostals and abdominal recti
 - e. Scaleni
 - f. Sternocleidomastoid muscles
49. The pleural pressure of a normal 56-year-old woman is approximately $-5 \text{ cm H}_2\text{O}$ during resting conditions immediately before inspiration (i.e., at functional residual capacity). What is the pleural pressure (in $\text{cm H}_2\text{O}$) during inspiration?
- a. +1
 - b. +4
 - c. 0
 - d. -3
 - e. -7

50. A patient has a dead space of 150 ml, functional residual capacity of 3 L, tidal volume of 650 ml, expiratory reserve volume of 1.5 L, total lung capacity of 8 L, and respiratory rate of 15 breaths/min. What is the residual volume?
- a. 500 ml
 - b. 1000 ml
 - c. 1500 ml
 - d. 2500 ml
 - e. 6500 ml

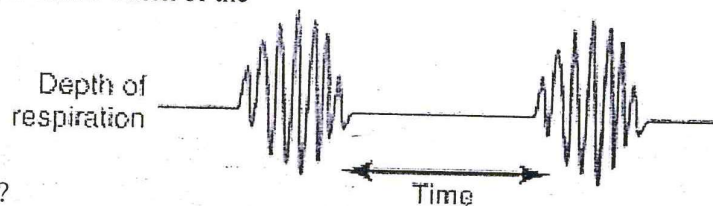
Questions 5 and 6



51. A 27-year-old man is breathing quietly. He then inhales as much air as possible and exhales as much air as he can, producing the spirogram shown in the previous figure. What is his expiratory reserve volume (in liters)?
- a. 2.0
 - b. 2.5
 - c. 3.0
 - d. 3.5
 - e. 4.0
 - f. 5.0
52. A 22-year-old woman inhales as much air as possible and exhales as much air as she can producing the spirogram shown in the figure. A residual volume of 1.0 L was determined using the helium dilution technique. What is her functional residual capacity (in liters)?
- a. 2.0
 - b. 2.5
 - c. 3.0
 - d. 3.5
 - e. 4.0
 - f. 5.0

53. A patient has a dead space of 150 ml, functional residual capacity of 3 L, tidal volume of 650 ml, expiratory reserve volume of 1.5 L, a total lung capacity of 8 L, respiratory rate of 15 breaths/min. What is the alveolar ventilation?
- 5 L/min
 - 7.5 L/min
 - 6.0 L/min
 - 9.0 L/min
54. The resistance of the pulmonary tree is so low that a 1 cm of water pressure gradient is sufficient to cause normal air flow during resting conditions. Which of the following often has a substantial resistance during pulmonary disease states that can limit alveolar ventilation?
- Alveoli
 - Bronchioles
 - Large bronchi
 - Small bronchi
 - Trachea
55. Which of the following will decrease pulmonary blood flow resistance?
- IV injection of Norepinephrine
 - Inhalation to total lung capacity
 - Breathing 5% O_2
 - Having the lung at FRC
56. A person with normal lungs at sea level (760 mm Hg) is breathing 50% oxygen. What is the approximate alveolar P_{O_2} ?
- 100
 - 159
 - 268
 - 330
 - 380
57. In which of the following conditions is alveolar P_{O_2} increased and alveolar P_{CO_2} decreased?
- Increased alveolar ventilation and unchanged metabolism
 - Decreased alveolar ventilation and unchanged metabolism
 - Increased metabolism and unchanged alveolar ventilation
 - Proportional increase in metabolism and alveolar ventilation
58. A 17-year-old female was bicycling without a helmet when she fell and hit her head. In the emergency room, she was not conscious and was receiving ventilator assistance. Her blood gases follow:
- $PaO_2 = 52$ mm Hg,
 $PaCO_2 = 75$ mm Hg, pH = 7.15, and $HCO_3^- = 31$ mM
 - The majority of the CO_2 was being transported as
 - CO_2 bound to plasma proteins
 - CO_2 bound to hemoglobin
 - Bicarbonate ions
 - Dissolved

59. The basic rhythm of respiration is generated by neurons located in the medulla. Which of the following limits the duration of inspiration and increases respiratory rate?
- Apneustic center
 - Dorsal respiratory group
 - Nucleus of the tractus solitarius
 - Pneumotaxic center
 - Ventral respiratory group
60. When respiratory drive for increased pulmonary ventilation becomes greater than normal, a special set of respiratory neurons that are inactive during normal quiet breathing then becomes active, contributing to the respiratory drive. These neurons are located in which of the following structures?
- Apneustic center
 - Dorsal respiratory group
 - Nucleus of the tractus solitarius
 - Pneumotaxic center
 - Ventral respiratory group
61. At a fraternity party a 17-year-old male places a paper bag over his mouth and breathes in and out of the bag. As he continues to breathe into this bag, his rate of breathing continues to increase. Which of the following is responsible for the increased ventilation?
- Increased alveolar PO_2
 - Increased alveolar PCO_2
 - Decreased arterial PCO_2
 - Increased pH
62. An anesthetized male is breathing with no assistance. He is then artificially ventilated for 10 min at his normal tidal volume but at twice his normal frequency. He is ventilated with a gas mixture of 60% O_2 and 40% N_2 . The artificial ventilation is stopped and he fails to breathe for several minutes. This apneic episode is due to which of the following?
- High arterial PCO_2 suppressing the activity of the peripheral chemoreceptors
 - Decrease in arterial pH suppressing the activity of the peripheral chemoreceptors
 - Low arterial PCO_2 suppressing the activity of the medullary chemoreceptors
 - High arterial PCO_2 suppressing the activity of the medullary chemoreceptors
 - Low arterial PCO_2 suppressing the activity of the peripheral chemoreceptors
63. The following diagram shows the depth of respiration of a 45-year-old man who suffered a head injury in an automobile accident. This "crescendo-decrescendo" pattern of breathing is called which of the

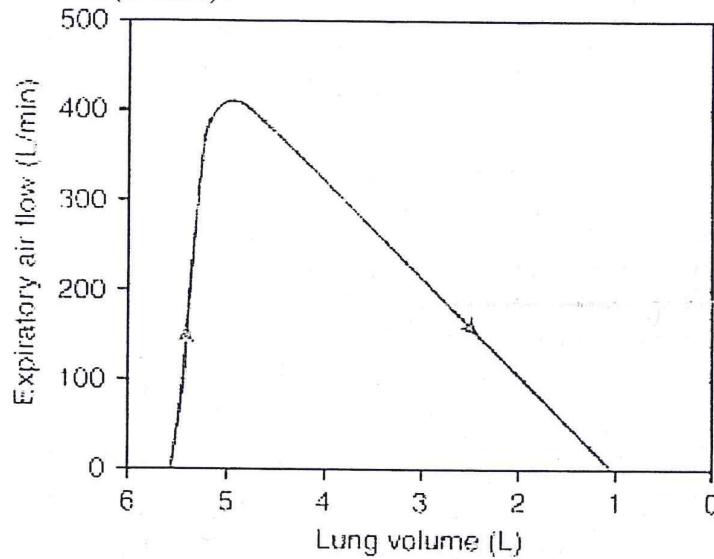


following?

- Apnea
- Biot breathing
- Cheyne-Stokes breathing

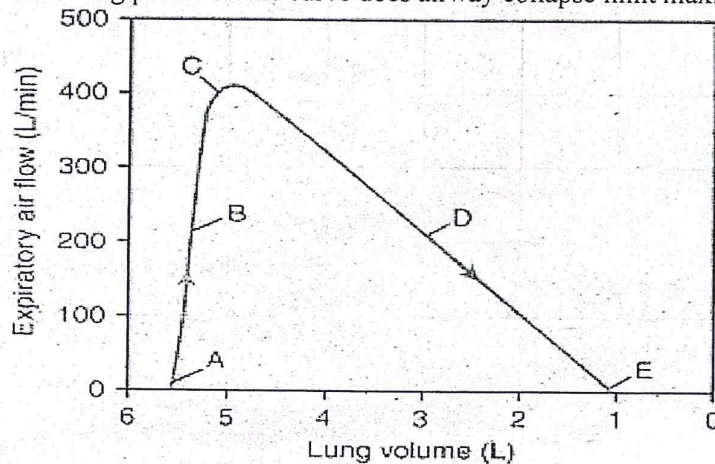
- d. Hyperpnea
- e. Tachypnea

64. A 45-year-old man inhaled as much air as possible and then expired with a maximum effort until no more air could be expired. This produced the maximum expiratory flow-volume curve shown in the following diagram. What is the forced vital capacity of this man (in liters)?



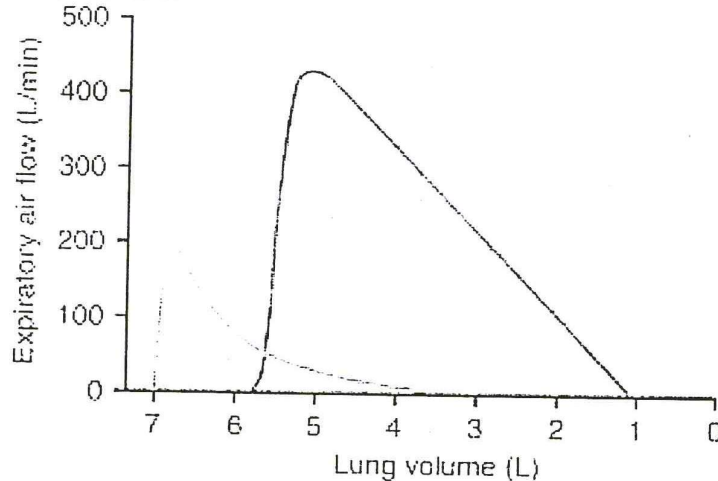
- a. 1.5
- b. 2.5
- c. 3.5
- d. 4.5
- e. 5.5
- f. 6.5

65. The maximum expiratory flow-volume curve shown in the following diagram is used as a diagnostic tool for identifying obstructive and restrictive lung diseases. At which of the following points on the curve does airway collapse limit maximum expiratory air flow?



66. The maximum expiratory flow-volume curves shown in the diagram below were obtained from a healthy individual (red curve) and a 57-year-old man who complains of

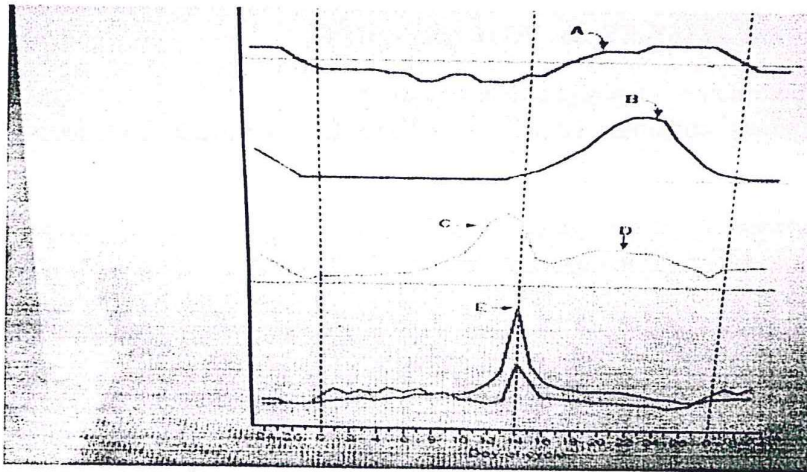
shortness of breath (green curve). Which of the following disorders is most likely present in the man?



- a. Asbestosis
 - b. Emphysema
 - c. Kyphosis
 - d. Scoliosis
 - e. Silicosis
 - f. Tuberculosis
67. Oxygen therapy is most beneficial in which of the following situations? Lung function is normal.
- a. Anemia
 - b. CO_2 retention (COPD)
 - c. Cyanide poisoning
 - d. High altitude
68. A person is planning to leaving Miami (sea level) and traveling to Colorado to climb Mt. Wilson, at 14,500 ft, barometric pressure = 450 mm Hg. He takes acetazolamide, a carbonic anhydrase inhibitor that forces the kidneys to excrete bicarbonate before his trip. Which of the following would be the expected response before he makes the trip?
- a. Alkalotic blood
 - b. Normal ventilation
 - c. Elevated ventilation
 - d. Normal arterial blood gases
69. A diver carries a 1000-L metal tank-box with an open bottom to a depth of 66 ft so that two divers can insert their heads and talk beneath the water. A person on the surface of the water pumps air into the box until the 1000-L box is completely filled with air. How much air from the surface is required to fill the box (in liters)?
- a. 1000
 - b. 2000
 - c. 3000
 - d. 4000
 - e. 5000
70. Which of the following functions of the Sertoli cells mediates negative feedback control of follicle-stimulating hormone (FSH) secretion?

- a) Synthesis of inhibin
- b) Synthesis of testosterone
- c) Aromatization of testosterone
- d) Maintenance of the blood–testes barrier

Use the diagram below to answer Q71-75



71. The increase shown at point A is caused by the effect of
 - a) Estrogen on the anterior pituitary
 - b) Progesterone on the hypothalamus
 - c) Follicle-stimulating hormone (FSH) on the ovary
 - d) Luteinizing hormone (LH) on the anterior pituitary
 - e) Prolactin on the ovary
72. Blood levels of which substance are described by curve B?
 - a) Estradiol
 - b) Estriol
 - c) Progesterone
 - d) Follicle-stimulating hormone (FSH)
 - e) Luteinizing hormone (LH)
73. The source of the increase in concentration indicated at point C is the
 - a) Hypothalamus
 - b) Anterior pituitary
 - c) Corpus luteum
 - d) Ovary
 - e) Adrenal cortex
74. The source of the increase in concentration at point D is the:
 - a) Ovary
 - b) Adrenal cortex
 - c) Corpus luteum
 - d) Hypothalamus
 - e) Anterior pituitary
75. The cause of the sudden increase shown at point E is:
 - a) Negative feedback of progesterone on the hypothalamus

- b) Negative feedback of estrogen on the anterior pituitary
- c) Negative feedback of follicle-stimulating hormone (fsh) on the ovary
- d) Positive feedback of fsh on the ovary
- e) Positive feedback of estrogen on the anterior pituitary

SECTION B: SHORT ANSWER QUESTIONS (55MIN)

76. Outline and briefly explain the types of circulatory shock. 25Min
77. Explain why diabetes is considered a prime risk factor for cardiovascular disease (CVD) 10Min
78. Outline the characteristics of normal Semen 5Min
79. Describe the main stages of spermatogenesis 15Min

SECTION C: LONG ESSAY QUESTIONS (50MIN)

80. Explain the hypothalamo-pituitary gonadal axis in regard to reproduction 25Min
81. 1 a.) Explain the shape of oxygen dissociation curve and the physiological importance of the shape 20Min.
82. b) List the factors that will shift hemoglobin dissociation curve to the left 5Min



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

**SECOND YEAR END YEAR EXAMINATION FOR THE DEGREE OF
BACHELOR OF MEDICINE AND SURGERY WITH INFORMATION
TECHNOLOGY**

MAIN CAMPUS – SPECIAL/SUPPLEMENTARY

MBS 201: MEDICAL PHYSIOLOGY II

Date: 11th January, 2018

Time: 9.00 - 12.00pm

INSTRUCTION:

- Follow the instructions as indicated in the question paper

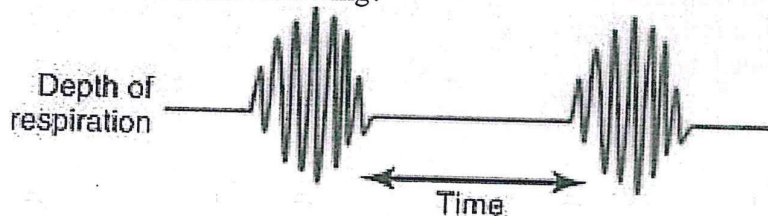


SECTION A: MULTIPLE CHOICE QUESTIONS (45MIN)
ONE CORRECT ANSWER

1. Oxygen therapy is most beneficial in which of the following situations if Lung function is normal?
 - A) Anemia
 - B) CO₂ retention (COPD)
 - C) Cyanide poisoning
 - D) High altitude
2. The total amount of O₂ carried by the circulation to the tissues/min. (oxygen delivery or total available oxygen)
 - A. Normally equals the rate of O₂ consumption by the body/min.
 - B. Is normally more than 95 per cent combined with haemoglobin.
 - C. Must fall by about half if haemoglobin concentration is halved.
 - D. Is more closely related to P_{O2} than to percentage saturation of the blood with O₂.
 - E. Must double if body oxygen consumption doubles.
3. Cyanosis:
 - A. May be caused by high levels of carboxyhaemoglobin in the blood.
 - B. May be caused by high levels of methaemoglobin in the blood.
 - C. Is seen in fingers of hands immersed in iced water.
 - D. Occurs more easily in anaemic than in polycythaemic patients.
 - E. Is severe in cyanide poisoning.
4. A patient with chronic respiratory failure
 - A. Shows an enhanced respiratory sensitivity to inhaled carbon dioxide.
 - B. Shows little or no respiratory response to hypoxia.
 - C. Is likely to have a low blood bicarbonate level.
 - D. Responds well when given 100 per cent oxygen to breathe.
 - E. Must have been breathing oxygen-enriched air if alveolar P_{CO2} is 150 mmHg (20 kPa).
5. Complete obstruction of a major bronchus usually results in
 - A. Collapse of the alveoli supplied by the bronchus.
 - B. A rise in local intrapleural pressure.
 - C. An increase in physiological dead space.
 - D. An increase in blood flow to the lung tissue supplied by the bronchus.
 - E. Cyanosis.
6. The following is false on a shift of the oxygen dissociation curve of blood to the left
 - A. Decreases the O₂ content of blood at a given P_{O2}.
 - B. Impairs O₂ delivery to the tissues at the normal tissue P_{O2}.
 - C. Occurs in blood perfusing cold extremities.
 - D. Occurs in blood stored for several weeks.
 - E. Is characteristic of fetal blood when compared with adult blood.
7. The work of breathing does not increase when
 - A. Lung compliance increases.
 - B. The subject exercises.



- C. The rate of breathing increases even though the minute volume stays constant.
 - D. The subject lies down.
 - E. Functional residual capacity increases.
8. The following diagram shows the depth of respiration of a 45-year-old man who suffered a head injury in an automobile accident. This "crescendo-decrescendo" pattern of breathing is called which of the following?



- A) Apnea
 - B) Biot breathing
 - C) Cheyne-Stokes breathing
 - D) Hyperpnea
 - E) Tachypnea
9. At a fraternity party a 17-year-old male places a paper bag over his mouth and breathes in and out of the bag. As he continues to breathe into this bag, his rate of breathing continues to increase. Which of the following is responsible for the increased ventilation?
- A. Increased alveolar P_{O_2}
 - B. Increased alveolar P_{CO_2}
 - C. Decreased arterial P_{CO_2}
 - D. Increased pH
10. At the onset of exercise, respiration rate and depth increase primarily because of
- A. increased blood carbon dioxide levels.
 - B. decreased blood oxygen levels
 - C. decreased blood pH levels.
 - D. input to the respiratory center from the cerebral motor cortex and proprioceptors.
11. Blood oxygen levels
- A. are more important than carbon dioxide levels in the regulation of respiration.
 - B. need to change only slightly to cause a change in respiration.
 - C. are detected by sensory receptors in the carotid and aortic bodies.
 - D. all of the above.
12. The chemosensitive area
- A. stimulates the respiratory center when blood carbon dioxide levels increase.
 - B. stimulates the respiratory center when blood pH increases.
 - C. is located in the pons.
 - D. stimulates the respiratory center when blood oxygen levels increase.
 - E. all of the above.
13. The alveolar ventilation is the:
- A. tidal volume times the respiratory rate

- B. minute ventilation plus the dead space.
 - C. amount of air available for gas exchange in the lungs.
 - D. vital capacity divided by the respiratory rate.
 - E. inspiratory reserve volume times minute ventilation.
14. Which of these parts of the brainstem is correctly matched with its main function?
- A. ventral respiratory groups—stimulate the diaphragm
 - B. dorsal respiratory groups—limit inflation of the lungs
 - C. pontine respiratory group—switch between inspiration and expiration
 - D. all of the above
15. Given these lung volumes:
- 1. tidal volume = 500 mL
 - 2. residual volume = 1000 mL
 - 3. inspiratory reserve volume = 2500 mL
 - 4. expiratory reserve volume = 1000 mL
 - 5. dead space = 1000 mL
- The vital capacity is"
- A. 3000 mL.
 - B. 3500 mL.
 - C. 4000 mL.
 - D. 5000 mL.
 - E. 6000 mL.
16. When blood passes through the tissues, the hemoglobin in blood is better able to combine with carbon dioxide because of the:
- A. Bohr effect.
 - B. Haldane effect.
 - C. chloride shift.
 - D. Boyle effect.
 - E. Dalton effect.
17. Carbon dioxide is mostly transported in the blood:
- A. dissolved in plasma.
 - B. bound to blood proteins
 - C. as HCO_3^- .
 - D. bound to the heme portion of hemoglobin.
 - E. bound to the globin portion of hemoglobin.
18. The lungs do not normally collapse because of
- A. surfactant.
 - B. pleural pressure.
 - C. elastic recoil.
 - D. both a and b.
19. During expiration, the alveolar pressure is:
- A. greater than the pleural pressure.
 - B. greater than the barometric pressure.
 - C. less than the barometric pressure.
 - D. unchanged.
20. The oxygen-hemoglobin dissociation curve is adaptive because it
- A. shifts to the right in the pulmonary capillaries and to the left in the tissue capillaries



- B. shifts to the left in the pulmonary capillaries and to the right in the tissue capillaries.
C. does not shift.
21. Oxygen is mostly transported in the blood
A. dissolved in plasma.
B. bound to blood proteins.
C. within HCO_3^- .
D. bound to the heme portion of hemoglobin.
22. The partial pressure of carbon dioxide in the venous blood is
A. greater than in the tissue spaces
B. less than in the tissue spaces
C. less than in the alveoli.
D. less than in arterial blood.
23. The rate of diffusion of a gas across the respiratory membrane increases as the:
A. respiratory membrane becomes thicker.
B. surface area of the respiratory membrane decreases.
C. partial pressure gradient of the gas across the respiratory membrane increases.
D. diffusion coefficient of the gas decreases.
E. e. all of the above.
24. During quiet expiration, the
A. abdominal muscles relax.
B. diaphragm moves inferiorly.
C. external intercostal muscles contract.
D. thorax and lungs passively recoil
E. all of the above.
25. If the total pressure of a gas is 760 mm Hg and its composition is 20% oxygen, 0.04% carbon dioxide, 75% nitrogen, and 5% water vapor, the partial pressure of oxygen is:
A. 15.2 mm Hg.
B. 20 mm Hg.
C. 118 mm Hg.
D. 152 mm Hg.
E. 740 mm Hg.
26. Immediately after the creation of an opening through the thorax into the pleural cavity:
A. air flows through the hole and into the pleural cavity.
B. air flows through the hole and out of the pleural cavity.
C. air flows neither out nor in.
D. the lung protrudes through the hole.
27. Compliance of the lungs and thorax
A. is the volume by which the lungs and thorax change for each unit change of alveolar pressure.
B. increases in emphysema.
C. decreases because of lack of surfactant.
D. all of the above.
28. Which muscle does not contract during forced expiration?
A. Internal intercostals
B. external intercostals



- C. Transversus abdominis
D. rectus abdominis
29. In A healthy individual with a total lung volume of 6 L, the amount of oxygen present in the lungs at the end of a tidal expiration is.
- A. 100ml
 - B. 210ml
 - C. 400ml
 - D. 1000ml
 - E. 2300ml
30. For a respiratory minute volume of 6L, which of the following combinations of breathing rate and tidal volumes allows for maximum alveolar ventilation in a normal individual?
- A. 10 breaths/min and 600mls
 - B. 15 breaths/min and 400mls
 - C. 20 breaths/min and 300mls
 - D. 30 breaths /min and 200mls
31. Physiologic dead space ventilation is decreased in or by?
- A. the upright position in comparison to the supine position.
 - B. positive pressure ventilation
 - C. neck flexion
 - D. emphysema
32. during a tidal respiration, airflow velocity is lowest in?
- A. nasopharynx
 - B. trachea
 - C. major bronchii
 - D. bronchioles
33. The major area of airway resistance during breathing is located in.
- A. oropharynx
 - B. trachea and large bronchi
 - C. intermediate sized bronchi
 - D. bronchioles <2mm in diameter.
34. Intrapleural pressure is positive during?
- A. deep inspiration
 - B. tidal expiration
 - C. forced expiration
 - D. tidal inspiration
35. Lung compliance is increased in
- A. pulmonary emphysema
 - B. pulmonary fibrosis
 - C. surfactant deficiency
 - D. acute pulmonary edema.
36. Dilation of bronchial smooth muscle is mediated by activation of
- A. alpha-1 receptors
 - B. alpha-2 receptors
 - C. M₂ receptors
 - D. β_2 adrenoceptors
37. Breathing in an individual with restrictive lung disease is typically



- A. slow and deep
- B. slow and shallow
- C. rapid and deep
- D. rapid and shallow

38. Which of the following statements is **incorrect**?

- A. In the upright position, ventilation is greater at the lung bases than the apex.
- B. in the upright position, lower lung zones are more compliant compared to upper lung zones.
- C. In the upright position, blood flow is greater at the lung bases than at the apices
- D. In the upright position, V/Q is normally highest in the lung apices
- E. positive intrapleural pressures during forced expiration prevent airways from closing

39. Despite a three to four fold increase in cardiac output, mean pulmonary artery pressure was observed to increase only by 20% during dynamic exercise. This is most likely due to:

- A. recruitment of collapsed pulmonary capillaries
- B. increase in left atrial pressure
- C. increase in sympathetic outflow
- D. increase in heart rate.

40. the hematocrit of venous blood is slightly greater than arterial blood because:

- A. the pH of venous blood is lower
- B. the addition of CO₂ increases osmoles in RBCs in venous blood.
- C. c. the exit of bicarbonate from RBCs increases water content of RBCs
- D. d. the chloride- bicarbonate exchanger is electroneutral.

41. The neurons pacing spontaneous breathing are located in:

- A. pneumotaxic center
- B. apneustic center
- C. c. pre-Bottzinger complex
- D. d. dorsal motor nucleus of vagus

42. What is/are the consequence(s) of a selective lesion of the pneumotaxic center?

- A. Respiratory frequency increases
- B. tidal volume decreases
- C. respiration becomes slower
- D. prolonged inspiratory spasms.

43. Administration of O₂ rich gas mixtures improves tissue oxygenation most in:

- A. hypoxic hypoxia
- B. anemic hypoxia
- C. hypokinetic hypoxia
- D. histotoxic hypoxia.

44. Pa O₂ is/ may be normal in tissue hypoxia due to:

- A. hypoventilation



- B. decompression at high altitude
- C. severe circulatory shock
- D. a reduction in diffusion capacity of lungs

A. anemia

45. hypocapnia is usually due to:

- A. hypoxemia
- B. alkalosis
- C. decreased CO₂ production
- D. alveolar hyperventilation

SECTION B: SAQs (30Mins)

1. Explain the regulation of:
 - a. Coronary circulation
 - b. Blood pressure

15Mins

15Mins

SECTION C: LEQ (115Mins)

1. Outline the hormonal changes that occur in puberty in males and females(20 minutes)
2. Describe the physiological changes that occur in the female reproductive organs during the menstrual cycle(25 minutes)
3. Describe the electrical and mechanical changes in the heart during a cardiac cycle
4. Describe the digestion and absorption of:
 - a. Lipids
 - b. Proteins

30Mins

15Mins

15Mins





MASENO UNIVERSITY
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**SECOND YEAR END OF YEAR EXAMINATION FOR THE DEGREE OF
BACHELOR OF MEDICINE AND BACHELOR OF SURGERY WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

MBS 202: MEDICAL BIOCHEMISTRY II

Date: 23rd November, 2017

Time: 2.00 - 5.00 pm

INSTRUCTIONS:

- The examination paper consists of THREE (3) Sections. (Total 140 mks)
- Answer ALL Questions in sections A,B and ANY TWO in section C.
- Section A has thirty (30) Multiple Choice Questions (MCQs) (60mks)
- Section B has nine (9) Short Answer Questions (SAQs) (50mks)
- Section C has three (3) Long Answer Questions (LEQs) (40mks)
- Answer questions in the Booklet provided. Begin each of the questions in section B and C in a new page of the answer booklet.

SECTION A: MULTIPLE CHOICE QUESTIONS**(60 MARKS)**

- 1). When two genes fail to assort independently, the term normally applied is;
A). Linkage
B). Recombination
C). Complementation
D). Mendelian inheritance
E). Incomplete dominance
- 2). Okazaki fragments are used to elongate;
A). Both strands in both directions
B). The leading strand toward the replication fork
C). The lagging strand toward the replication fork
D). The leading strand away from the replication fork
E). The lagging strand away from the replication fork
- 3). Transcription is initiated when RNA polymerase binds to;
A). A codon
B). An initiator
C). A promoter
D). An enhancer
E). A transcriptor
- 4). DNA replication is called semiconservative because _____ of the original duplex appears in the duplex formed in replication.
A). All
B). Half
C). Most
D). None
E). Hardly any
- 5). Amino acid catabolism predominantly occurs in the;
A). Liver
B). Brain
C). Kidney
D). Adipose tissue
E). Cardiac muscle
- 6). Cri-du-chat syndrome is associated with;
A). A missing chromosome number 5
B). An additional chromosome number 5
C). A deletion of the long arm of chromosome number 5
D). A deletion of the short arm of chromosome number 5
E). Duplication of both the short and long arm of chromosome number 5
- 7). In the process of translation;
A). Nucleotide sequences of tRNA are established
B). A polypeptide is formed in response to the rRNA nucleotide sequence
C). rRNA is synthesized with sequences complementary to those of tRNA
D). A polypeptide is formed as dictated by the nucleotide sequence in mRNA



- E). A strand of mRNA is formed with nucleotide sequences complementary to those of DNA
- 8). Which of the following statement is true about prions?
- A). Highly immunogenic
 - B). May be readily cultured
 - C). Have long incubation periods
 - D). Contain DNA, RNA and proteins
 - E). Readily inactivated by autoclave at 121°C
- 9). Which one of the following processes does NOT occur in a cell stimulated by a steroid hormone?
- A). The hormone binds to a receptor molecule in the cytoplasm
 - B). The steroid hormone enters the cell by crossing the plasma membrane
 - C). DNA is transcribed, mRNA is translated, and the result is protein synthesis
 - D). The hormone-receptor complex binds the chromatin and activates certain genes
 - E). The second messenger cyclic AMP is stimulated by the hormone-receptor complex
- 10). Transcription factors appear to be unable to bind to a nucleosome because;
- A). Of inhibition of RNA polymerase
 - B). Of histones positioned over promoters
 - C). Activators are inhibited by the configuration
 - D). Operators are placed in an inaccessible position
 - E). Nucleosomes are especially vulnerable to repressors
- 11). The role of methylation of DNA is now viewed as;
- A). Interfering with DNA transcription by blocking base pairing between cytosine and guanine
 - B). Complexing with enhancers to prevent transcription
 - C). Insuring that genes that are turned off, stay off
 - D). Irrelevant to gene transcription
 - E). Prevention of mutation
- 12). Which of the following states the proper order of phases in the life cycle of animal viruses;
- A). Adsorption, penetration, synthesis, uncoating, assembly and release
 - B). Adsorption, penetration, assembly, synthesis, uncoating, and release
 - C). Adsorption, uncoating, penetration, synthesis, assembly, and release
 - D). Adsorption, penetration, uncoating, synthesis, release, and assembly
 - E). Adsorption, penetration, uncoating, synthesis, assembly, and release
- 13). The genotypes of a husband and wife are $I^A I^B \times I^A i$. Among the blood types of their children, how many different genotypes and phenotypes are possible?
- A). 2 genotypes; 3 phenotypes
 - B). 3 genotypes; 4 phenotypes
 - C). 4 genotypes; 4 phenotypes
 - D). 3 genotypes; 3 phenotypes
 - E). 4 genotypes; 3 phenotypes
- 14). Which of the following statements about the transport of substances through a cell membrane is correct?



- A). Small polar molecules require specific transport systems for passage across a membrane
 - B). Small inorganic ions can diffuse through a cell membrane along a concentration gradient
 - C). Small polar molecules can diffuse through a cell membrane along a concentration gradient
 - D). Small polar molecules cannot be transported through a cell membrane against a concentration gradient
 - E). Small inorganic ions use the process of endocytosis leading to formation of vesicles which then exclude its contents and deposits them outside the side
- 15). All of the following substances serve as carrier types at the blood brain barrier EXCEPT;
- A). Choline carriers
 - B). Basic amino acid carriers
 - C). Vitamin and lipid carriers
 - D). Monocarboxylic acid carriers
 - E). Large neutral amino acid carriers
- 16). Suppose a certain gene contains the double-stranded sequence:
 5'- ATGTTTAGCGCC -3'
 3'- TACAAATCGCGG -5'
 If the top strand is the sense strand and codes for a mRNA whose sequence begins 'ATG', which of the following would be the sequence of the corresponding segment of antisense RNA?
- A). 5'-AUGUUUAGCGCC-3'
 - B). 5'-CCGCGAUUUGUA-3'
 - C). 5'-GGCGCUAAACAU-3'
 - D). 5'-UACAAAUCGCGG-3'
 - E). 3'-GGCGCUAAACAU-5'
- 17). Which statement about the elongation of the lagging strand during DNA replication is correct?
- A). It progresses (grows) toward the replication fork
 - B). It requires a short RNA primer to proceed
 - C). It is synthesized in a 3' → 5' direction
 - D). It is synthesized by DNA ligase
 - E). It is synthesized continuously
- 18). Tumor-suppressor genes;
- A). Encode proteins that prevent binding of cyclins
 - B). Are involved in the cellular response to EGF
 - C). Includes the widely studied myc gene
 - D). Stimulates the binding of GTP
 - E). Are genetically recessive
- 19). Cancer is often the result of activation of _____ to _____ and the inactivation of _____ genes.
- A). Proto-suppressor genes, suppressors, oncogenes
 - B). Proto-oncogenes, oncogenes, tumor-suppressor genes
 - C). Oncogenes, tumor-suppressor genes, proto-oncogenes

